HARD WIRED OUTPUTS

The hard wired outputs of the EMIC include the following:

- **Courtesy Lamp Feed**
- **Axle Lock Switch Enable 1 (w/Off-Road Package Only)**
- **Axle Lock Switch Enable 2 (w/Off-Road Package Only)**
- **Panel Lamp Feed**
- **Rear Window Defogger Relay Control**

Refer to the appropriate wiring information for additional details.

GROUNDS

The EMIC receives a ground path through the following hard wired circuit:

- **Ground - G201**

Refer to the appropriate wiring information for additional details.

COMMUNICATION

The EMIC has the following data bus communication circuit:

- **PCI Data Bus**

Refer to the appropriate wiring information for additional details.

DIAGNOSIS AND TESTING - INSTRUMENT CLUSTER

If all of the instrument cluster gauges and/or indicators are inoperative, refer to PRELIMINARY DIAGNOSIS. If an individual gauge or Programmable Communications Interface (PCI) data bus message-controlled indicator is inoperative, refer to ACTUATOR TEST. If an individual hard wired indicator is inoperative, refer to the diagnosis and testing information for that specific indicator. If the instrument cluster interior lighting control function is inoperative, refer to INTERIOR LIGHTING CONTROL DIAGNOSIS. If the axle locker control is inoperative, refer to AXLE LOCKER CONTROL DIAGNOSIS. If the instrument cluster rear window defogger control function is inoperative, refer to REAR WINDOW DEFOGGER CONTROL DIAGNOSIS.

Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

CAUTION: Instrument clusters used in this model automatically configure themselves for compatibility with the features and optional equipment in the vehicle in which they are initially installed. The instrument cluster is programmed to do this by embedding the Vehicle Identification Number (VIN) and other information critical to proper cluster operation into electronic memory. This embedded information is learned through electronic messages received from other electronic modules in the vehicle over the Programmable Communications Interface (PCI) data bus, and through certain hard wired inputs received when the cluster is connected to the vehicle electrically. Once configured, the instrument cluster memory may be irreparably damaged and certain irreversible configuration errors may occur if the cluster is connected electrically to another vehicle; or, if an electronic module from another vehicle is connected that provides data to the instrument cluster (including odometer values) that conflicts with that which was previously learned and stored. Therefore, the practice of exchanging (swapping) instrument clusters and other electronic modules in this vehicle with those removed from another vehicle must always be avoided. Failure to observe this caution may result in instrument cluster damage, which is not reimbursable under the terms of the product warranty. Service replacement instrument clusters are provided with the correct VIN, and the certified odometer value embedded into cluster memory, but will otherwise be automatically configured for compatibility with the features and optional equipment in the vehicle in which they are initially installed.

NOTE: Certain indicators in this instrument cluster are automatically configured. This feature allows those indicators to be activated for deactivated for compatibility with certain optional equipment. If the problem being diagnosed involves improper illumination of the airbag indicator, the cruise indicator, or the SKIM indicator when the vehicle does not have this equipment, the instrument cluster must be replaced with a new unit.

PRELIMINARY DIAGNOSIS

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.